

Strategic coexistence theory for evolutionary game theory

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Abstract

Introduction

Understanding the mechanisms that promote the evolutionary stability of competing strategies is a fundamental aim in evolutionary biology (Lewontin, 1961; Slobodkin, 1964; Smith and Price, 1973; Taylor and Jonker, 1978). To this end, evolutionary game theory (EGT) and replicator dynamics have provided key foundations, allowing us to predict the outcome of strategic competition through payoff inequalities, invasion criteria, and stability analysis (Hofbauer and Sigmund, 1998; Cressman and Tao, 2014). However, less attention has been given to similarities between replicator dynamics and community ecology until recently (Page and Nowak, 2002; Tarnita and Traulsen, 2025; Allesina, 2026), even though the classic replicator equation stems from the generalized Lotka-Volterra equation.

Interestingly, even a simple two-strategy game can yield heterogeneous outcomes that mirror outcomes in community ecology: the dominance of a singular strategy (prisoner’s dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). Such differences in conditions that promote evolutionary stability and dominance of a singular strategy and those that permit strategy coexistence indicate that EGT can be understood from the perspective of coexistence theory. Recasting game dynamics as a coexistence problem immediately raises the following question: when two strategies coexist, does the corresponding mechanism stabilize or equalize competition? Even though EGT allows us to predict whether two strategies can coexist or not, a framework for quantifying coexistence mechanism between competing strategies remains elusive.

Modern coexistence theory (MCT) from community ecology presents a particularly useful framework for quantifying coexistence mechanisms. MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to be greater than average fitness differences. In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor. In EGT, a strategy’s niche can be interpreted as strategic environments defined by the composition of competing strategies and the resulting interactions. Thus, stabilizing niche differences arise when each strategy is favored by a different

strategic environment, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast, fitness differences represent differences in the ability of each strategy to compete under the limiting strategic environments created by its competitor. This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another.

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, especially for studying pathogen strain competition, allowing one to directly quantify niche and fitness differences of competing entities, be they species or strains. Inspired by this effort, we present a Strategic Coexistence Theory (SCT) for predicting coexistence and exclusion of competing strategies in EGT by quantifying strategic niche and fitness differences. We begin by introducing SCT, which can be applied to any generic system of EGT described by replicator equations. We show that SCT can recover outcomes from classic, two strategy games, providing reinterpretation of classic results from coexistence-theoretic perspective. We then apply SCT to comparing five mechanisms for the evolution of cooperation, revealing that direct and indirect reciprocity destabilizes competition between cooperation and defection, thus pushing the system towards priority effect regime. Finally, we consider a case study of microbial cooperation, where nonlinear microbial growth can both stabilize and equalize competition between cooperation and defection, allowing coexistence of strategies.

Strategic coexistence theory

We begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs (Figure 1):

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where x_i represents the frequency of strategy i , $\pi_i(\mathbf{x})$ represents the payoff of strategy i for a given strategy distribution $\mathbf{x}$, and $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$ represents the average payoff. Since strategy frequencies must sum to 1 (i.e., $\sum_i x_i = 1$), a single-strategy equilibrium dominated by strategy i can be described by a unit vector $\mathbf{e}_i$, where all elements are equal to zero except for the i -th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies i and j . However, because payoffs in many games can be negative, payoff ratios are difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \quad (2)$$

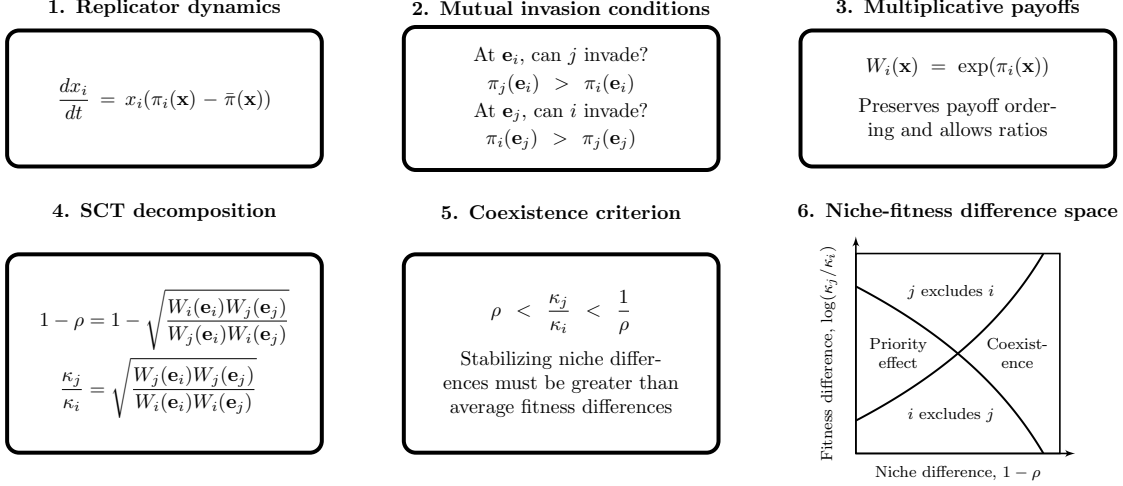


Figure 1: **A schematic diagram summarizing strategic coexistence theory.** Strategic coexistence theory decomposes pairwise replicator dynamics into stabilizing and equalizing mechanisms. Mutual invasion conditions can be rewritten on a multiplicative payoff scale, allowing us to derive strategic niche difference $1 - \rho$ and fitness difference κ_j/κ_i . Then, the inequality $\rho < \kappa_j/\kappa_i < 1/\rho$ determines mutual invasion.

For convenience, we refer to $W_i(\mathbf{x})$ as the multiplicative payoff of strategy i .

First, we define the niche overlap ρ between strategies i and j as the ratio between the geometric mean of each strategy's multiplicative payoff in the strategic environment that it creates when common, $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$, and the geometric mean of each strategy's multiplicative payoff in the environment generated by its competitor, $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$. The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (3)$$

When $1 - \rho > 0$, each strategy performs better, on average, in the environment generated by its competitor than in the environment generated by itself. In other words, intraspecific competition is stronger than interspecific competition.

Second, we define the fitness difference as the ratio between the geometric means of the multiplicative payoff of each strategy across the two single-strategy environments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (4)$$

In other words, strategy j has an average competitive advantage over strategy i ($\kappa_j/\kappa_i > 1$) when it performs better, on average, across the environments generated by itself and its competitor. Then, analogous to MCT, mutual invasion requires the

89 fitness ratio to fall within the bounds set by niche overlap:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (5)$$

90 This decomposition allows us to understand how different mechanisms contribute to
91 the coexistence or exclusion of competing strategies.

92 **Classic two-strategy games**

93 As an illustrative example, we begin with classic two-strategy games, which can be
94 described by the following payoff matrix (Figure 2A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (6)$$

95 Here, the first and second rows represent strategies 1 and 2, respectively, and the
96 columns represent the strategy of the interacting opponent. Writing x_1 to denote
97 the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (7)$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (8)$$

98 Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1 - x_1) [(R - T)x_1 + (S - P)(1 - x_1)]. \quad (9)$$

99 The signs of $T - R$ and $S - P$ determine whether each strategy can invade the
100 other when rare. Specifically, strategy 2 can invade strategy 1 when $T > R$, whereas
101 strategy 1 can invade strategy 2 when $S > P$. These two invasion conditions partition
102 the game space into four regimes (Figure 2B): prisoner's dilemma ($T > R$, $P >$
103 S), hawk-dove game ($T > R$, $S > P$), stag-hunt game ($R > T$, $P > S$), and
104 harmony game ($R > T$, $S > P$). In standard EGT, these outcomes are obtained by
105 analyzing the stability of equilibrium conditions. For simple games, this analysis is
106 straightforward.

107 A coexistence-theoretic approach recovers the same classification but provides
108 a different interpretation (Figure 2C). For a two-strategy game determined by this
109 payoff matrix, the niche difference and fitness difference are given by

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \quad (10)$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (11)$$

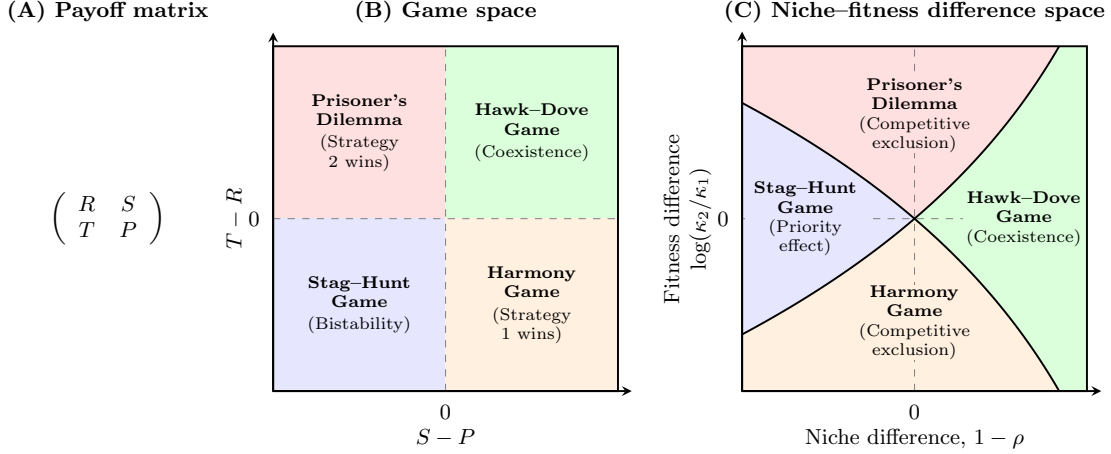


Figure 2: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences $T - R$ and $S - P$. Prisoner's dilemma and harmony games lead to the dominance of one strategy; hawk-dove games allow coexistence; and stag-hunt games result in bistability. (C) Mapping of the same four games onto niche-fitness difference space. Hawk-dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner's dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag-hunt games correspond to the priority-effect regime.

As shown above, mutual invasion requires $\rho < \kappa_2/\kappa_1 < 1/\rho$, which is equivalent to $T > R$ and $S > P$. This corresponds to the hawk-dove game, in which each strategy can invade when rare and the two strategies coexist. The remaining games also have natural interpretations in niche-fitness difference space. Prisoner's dilemma and harmony games represent competitive exclusion regimes, where one strategy has a fitness advantage that exceeds niche differences. In contrast, stag-hunt games represent a priority-effect regime, where neither strategy can invade the other and the outcome depends on initial conditions. Therefore, coexistence theory not only allows us to recover the classic results from two-strategy games but also provides new interpretations for these results in terms of niche and fitness differences. Building on this, we next apply this framework to comparing core mechanisms of cooperation.

121 Comparisons of mechanisms for the evolution of co- 122 operation

123 Even though cooperative behavior can be found across a wide range of biological
124 systems, a simple prisoner’s dilemma shows that cooperation is not an evolutionarily
125 stable strategy. Thus, many researchers have sought to identify mechanisms that
126 promote evolutionary stability of cooperation. Here, we compare five classic mecha-
127 nisms for the evolution of cooperation using SCT and ask how they differ from the
128 perspective of strategy coexistence.

129 Specifically, we consider the following mechanisms: kin selection, direct reci-
130 procity, indirect reciprocity, network reciprocity, and group selection (Figure 3A).
131 Kin selection favors cooperation when two interacting individuals are genetic rela-
132 tives, where the genetic relatedness r must be greater than the cost-to-benefit ratio:
133 $r > c/b$. Direct reciprocity favors cooperation when current cooperation may pro-
134 mote future cooperation by the same individual through repeated encounters. In
135 this case, the probability of repeated interaction (i.e., “next round”) must be greater
136 than the cost-to-benefit ratio: $w > c/b$. Indirect reciprocity favors cooperation when
137 current cooperation may promote future cooperation by different individuals through
138 reputation. Thus, the social acquaintanceship (i.e., the probability of knowing some-
139 one’s reputation) must be greater than the cost-to-benefit ratio: $q > c/b$. Network
140 reciprocity favors cooperation via clustering of cooperators, requiring the benefit-to-
141 cost ratio to be greater than the average number of neighbors: $b/c > k$. Finally,
142 group selection favors cooperation via multilevel selection, where groups with more
143 cooperators exhibit faster growth at the group level even if defectors reproduce faster
144 at the individual level. This requires $b/c > 1 + (n/m)$, where n is the maximum group
145 size and m is the number of groups. All of these mechanisms can be described by
146 a simple 2×2 payoff matrix (Figure 3A), from which the corresponding niche and
147 fitness differences can be computed using Eq. (10) and Eq. (11).

148 First, the prisoner’s dilemma provides a baseline scenario, where defection is
149 evolutionarily stable (Figure 3A). From the perspective of SCT, there is complete
150 niche overlap, and therefore defection competitively excludes cooperation (Figure
151 3B). Building on this, SCT shows that five mechanisms promote evolutionary sta-
152 bility of cooperation via different dynamical routes (Figure 3B). For example, kin
153 selection allows the fitness of cooperation to increase as genetic relatedness r increases
154 (Figure 3B). However, it does not affect the niche difference between cooperation and
155 defectors. Similarly, network reciprocity and group selection modulates the fitness
156 of cooperation without causing any changes in strategic niches. In contrast, direct
157 and indirect reciprocity destabilizes competition while increasing the fitness of coop-
158 eration, thereby pushing the competition outcome towards a priority effect regime.
159 Equivalently, these two mechanisms promote bistability of defection and cooperation.

160 SCT further reveals another difference between direct and indirect reciprocity.
161 In particular, direct reciprocity is unable to generate sufficient fitness difference to
162 competitively exclude defection even when the probability of next round w reaches 1.

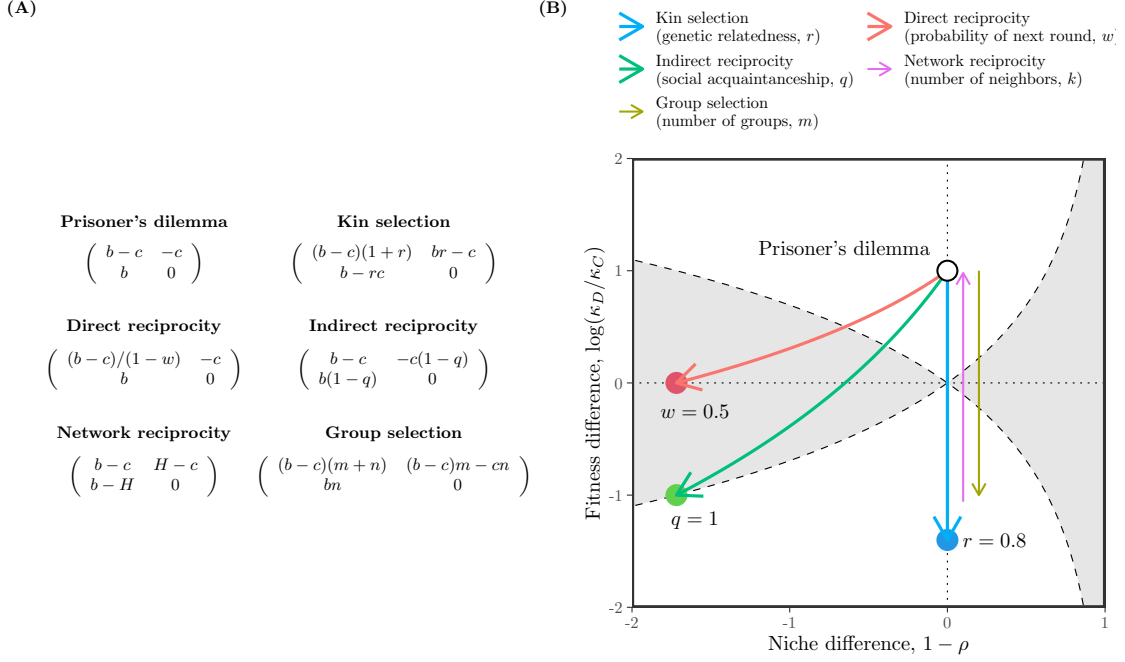


Figure 3: **Comparisons of five mechanisms for the evolution of cooperation using SCT.** (A) Payoff matrices for the baseline scenario (prisoner's dilemma) and five mechanisms for the evolution of cooperation (Nowak, 2006). Prisoner's dilemma is characterized by the benefit for the recipient b and the cost for the donor c . (B) Changes in niche and fitness differences driven by each mechanism. Each arrow represents the direction of change as we increase the corresponding parameter, indicated in figure legends. Note that kin selection, network reciprocity, and group selection all results in changes in fitness difference without causing any changes in niche difference. Therefore, we indicate the direction of change for network reciprocity and group selection separately to avoid overlapping arrows.

163 In contrast, indirect reciprocity is able to generate sufficient fitness difference to push
 164 the system towards the boundary between priority-effect regime and competitive-
 165 exclusion regime as social acquaintanceship q reaches 1, indicating that an additional
 166 mechanism will be able to promote complete dominance of cooperation.

167 We note that none of these mechanisms allow coexistence of cooperation and
 168 defection. Therefore, we move on to the final example, applying SCT to micro-
 169 bial cooperation, where nonlinear benefits and environmental context can promote
 170 coexistence of cooperators and defectors.

171 Microbial cooperation and public goods

172 Here, we consider a simple, phenomenological model of competition between cooper-
 173 ating and cheating yeast in a cooperative sucrose-metabolism system, where invertase
 174 production by cooperators allows cheaters to utilize glucose, a public good released
 175 through sucrose hydrolysis (Gore et al., 2009). Specifically, invertase production by
 176 cooperators incurs a cost c and generates a total benefit of 1 that is captured pri-
 177 vately with efficiency ϵ . Assuming a linear relationship between glucose availability
 178 and microbial growth, the payoffs for cooperators π_C and cheaters (defectors) π_D can
 179 then be written as

$$\pi_C(f) = \epsilon + f(1 - \epsilon) - c, \quad (12)$$

$$\pi_D(f) = f(1 - \epsilon), \quad (13)$$

180 where the use of glucose depends on the fraction of cooperators f . The dynamics of
 181 cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\pi_C(f) - \pi_D(f)] \quad (14)$$

$$= f(1 - f)(\epsilon - c) \quad (15)$$

182 For this linear growth model, the payoff difference is fixed to $\epsilon - c$ independent of
 183 f (Figure 4A). Therefore, cooperation is favored when efficiency is greater than the
 184 cost of cooperation, $\epsilon > c$, whereas defection is favored when the cost is greater than
 185 efficiency, $c > \epsilon$ (Gore et al., 2009).

186 Our framework allows us to reinterpret this result from the perspective of coexis-
 187 tence theory. In particular, the two strategies have no niche difference, corresponding
 188 to complete niche overlap, $\rho = 1$. In contrast, the fitness difference is determined by
 189 the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \quad (16)$$

190 Therefore, this model always predicts competitive exclusion, except in the neutral
 191 case $c = \epsilon$. The outcome is determined entirely by the fitness difference between
 192 cooperators and defectors, rather than by stabilizing niche differences. This example
 193 resembles the prisoner's dilemma discussed in the previous example (Figure 3).

194 In real biological systems, the effect of glucose on microbial growth can follow a
 195 nonlinear relationship:

$$\pi_C(f) = [\epsilon + f(1 - \epsilon)]^\alpha - c, \quad (17)$$

$$\pi_D(f) = [f(1 - \epsilon)]^\alpha, \quad (18)$$

196 where $\alpha = 0.15$ reflects the experimental relationship between glucose and microbial
 197 growth (Gore et al., 2009). Then, the dynamics of cooperators and defectors are

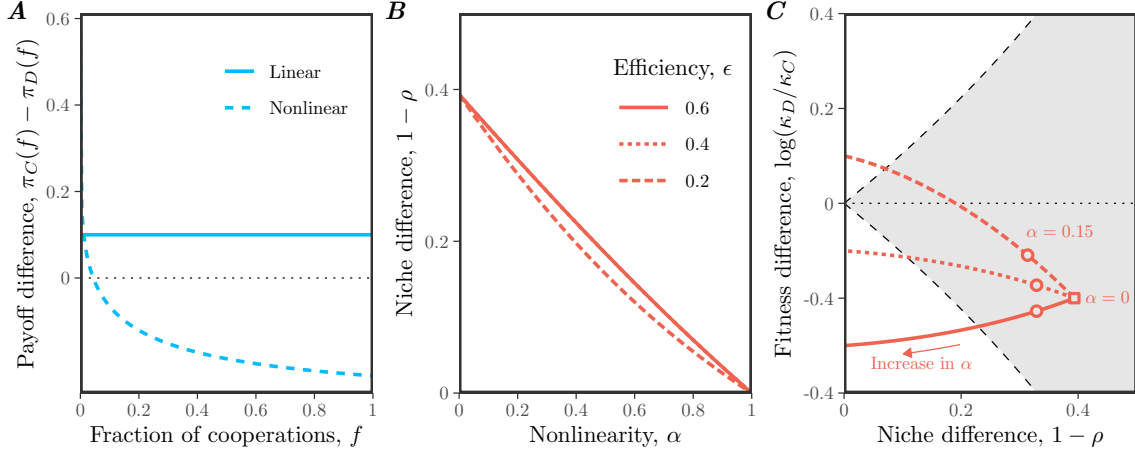


Figure 4: **Effects of nonlinear microbial growth on coexistence of cooperation and defection.** (A) Payoff differences for linear and nonlinear models of microbial growth. In panel A, we assumed $\epsilon = 0.4$, $c = 0.3$, and $\alpha = 0.15$ for illustrative purposes. (B) Effect of nonlinearity α on niche difference $1 - \rho$. Each line assumes a different value of efficiency ϵ . Note that lines for $\epsilon = 0.6$ and $\epsilon = 0.4$ are completely overlapping. (C) Effect of nonlinearity α on both niche difference $1 - \rho$ and fitness difference $\log(\kappa_D/\kappa_C)$. The gray shaded area represents the coexistence regime. The square represents the condition when $\alpha = 0$. The circles represent the condition when $\alpha = 0.15$, which correspond to the experimental relationship between glucose and microbial growth (Gore et al., 2009). In panels B and C, we assumed $c = 0.3$, while allowing ϵ and α to vary.

described by the following replicator equation:

$$\frac{df}{dt} = f(1-f) [\{\epsilon + f(1-\epsilon)\}^\alpha - c - f(1-\epsilon)^\alpha]. \quad (19)$$

Under this nonlinear assumption, the payoff difference decreases with the cooperator fraction f , which can promote the coexistence of cooperators and defectors (Figure 2A): cooperators have a competitive advantage when rare, whereas defectors have a competitive advantage when cooperators are abundant.

From a coexistence-theoretic perspective, this means that nonlinear growth can generate sufficient stabilizing niche difference to overcome the fitness difference. In particular, we can show that the niche and fitness differences for this model are given by

$$1 - \rho = 1 - \exp\left(\frac{1 - \epsilon^\alpha - (1 - \epsilon)^\alpha}{2}\right), \quad (20)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1 - \epsilon)^\alpha - 1 - \epsilon^\alpha}{2}\right). \quad (21)$$

These expressions clarify the separate roles of cost, efficiency, and nonlinearity. For example, the cost of cooperation does not affect niche difference. Instead, niche

209 difference is generated by the concave relationship between glucose availability and
 210 microbial growth, which increases from 0 to $1 - \exp(-1/2)$ as α goes from 1 to 0
 211 (Figure 4B,C). In contrast, as $\alpha \rightarrow 0$, the fitness difference approaches

$$\log(\kappa_D/\kappa_C) \rightarrow c - \frac{1}{2}, \quad (22)$$

212 which no longer depends on the private capture efficiency ϵ (Figure 4C). Thus, strong
 213 concavity not only generates stabilizing niche differences, but also acts as an equal-
 214 izing mechanism (Figure 4C).

215 Discussion

216 Standard evolutionary game theory and replicator equations provide powerful tools
 217 for predicting the outcome of strategy competition, including strategic dominance,
 218 coexistence, and bistability. However, these outcomes are typically understood by
 219 analyzing payoff differences within individual games, which makes it difficult to com-
 220 pare coexistence mechanisms across models. Building on Chesson’s Modern Coexis-
 221 tence Theory from community ecology, we developed a strategic coexistence theory
 222 (SCT) that allows us to decompose strategy competition in terms of stabilizing niche
 223 differences and fitness differences. This decomposition provides a unifying framework
 224 for comparing different games from the perspective of coexistence theory, allowing
 225 us to identify stabilizing and equalizing mechanisms for strategy competition.

226 To demonstrate the utility of SCT, we considered three case studies. First, the
 227 analysis of classic two-strategy games showed that SCT recovers the original clas-
 228 sification for these classic games, highlighting the similarities between evolutionary
 229 games and community ecology. Then, we compared different mechanisms for the evo-
 230 lution of cooperation, which showed that each mechanism promotes cooperation via
 231 different dynamical routes. Kin selection, network reciprocity, and group selection
 232 primarily affects the fitness difference between cooperation and defection, whereas di-
 233 rect and indirect reciprocity can destabilize competition and generate priority effects.
 234 Finally, the analysis microbial cooperation revealed that

Supplementary Text

S1 Derivation of the strategic coexistence theory

To derive strategic coexistence theory (SCT), we begin with replicator equations:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (\text{S1})$$

where x_i represents the frequency of strategy i , π_i represents the payoff of strategy i , and $\bar{\pi}$ represents the average payoff. Then, the mutual invasion condition for two competing strategies i and j can be written as

$$\pi_j(\mathbf{e}_j) < \pi_i(\mathbf{e}_j), \quad (\text{S2})$$

$$\pi_i(\mathbf{e}_i) < \pi_j(\mathbf{e}_i). \quad (\text{S3})$$

These inequalities will be preserved when we exponentiate the payoffs, $W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x}))$:

$$W_j(\mathbf{e}_j) < W_i(\mathbf{e}_j), \quad (\text{S4})$$

$$W_i(\mathbf{e}_i) < W_j(\mathbf{e}_i). \quad (\text{S5})$$

Rearranging, we obtain the following inequality:

$$\frac{W_i(\mathbf{e}_i)}{W_j(\mathbf{e}_i)} < 1 < \frac{W_i(\mathbf{e}_j)}{W_j(\mathbf{e}_j)}. \quad (\text{S6})$$

By multiplying $\sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$ on both sides, we obtain:

$$\sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}} \quad (\text{S7})$$

Thus, by defining niche overlap ρ and fitness difference κ_j/κ_i as follows,

$$\rho = \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S8})$$

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S9})$$

we obtain the following inequality that defines conditions for mutual invasion and long-term coexistence of two competing strategies:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S10})$$

248 S2 Applications to classic two strategy games

249 Consider a two strategy game described by the following payoff matrix:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (\text{S11})$$

250 There are four classic games that we can define from this simple payoff matrix:
 251 prisoner's dilemma ($T > R > P > S$), hawk-dove game ($T > R, S > P$), stag-hunt
 252 game ($R > T, P > S$), and harmony game ($R > T, S > P$). Here, we show that
 253 each game corresponds to a distinct region in the niche-fitness difference space under
 254 SCT.

255 To do so, we first begin by deriving expressions for niche overlap ρ and fitness
 256 difference κ_j/κ_i . Writing x_1 to denote the frequency of strategy 1, the expected
 257 payoff for each strategy corresponds to:

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (\text{S12})$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (\text{S13})$$

258 Then, multiplicative payoffs for each strategy when rare can be written as:

$$W_1(1) = \exp(R) \quad (\text{S14})$$

$$W_1(0) = \exp(S) \quad (\text{S15})$$

$$W_2(1) = \exp(T) \quad (\text{S16})$$

$$W_2(0) = \exp(P) \quad (\text{S17})$$

259 Therefore, we have:

$$\rho = \exp\left(\frac{R + P - S - T}{2}\right), \quad (\text{S18})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (\text{S19})$$

260 As before, SCT allows us to predict the mutual invasion of two strategy using the
 261 following inequality:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S20})$$

262 Then, simple algebra reveals that each condition in mutual invasion inequality
 263 can be equivalently described by the payoff inequality:

$$\rho < \frac{\kappa_2}{\kappa_1} \iff R < T \quad (\text{S21})$$

$$\rho > \frac{\kappa_2}{\kappa_1} \iff R > T \quad (\text{S22})$$

$$\frac{\kappa_2}{\kappa_1} < \frac{1}{\rho} \iff P < S \quad (\text{S23})$$

$$\frac{\kappa_2}{\kappa_1} > \frac{1}{\rho} \iff P > S \quad (\text{S24})$$

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